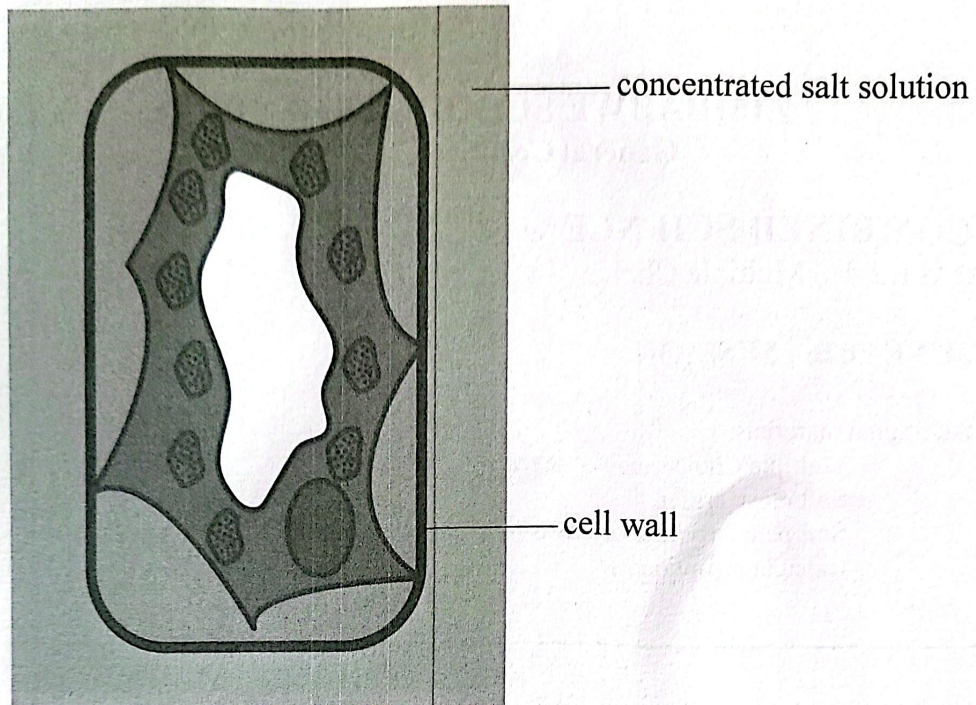


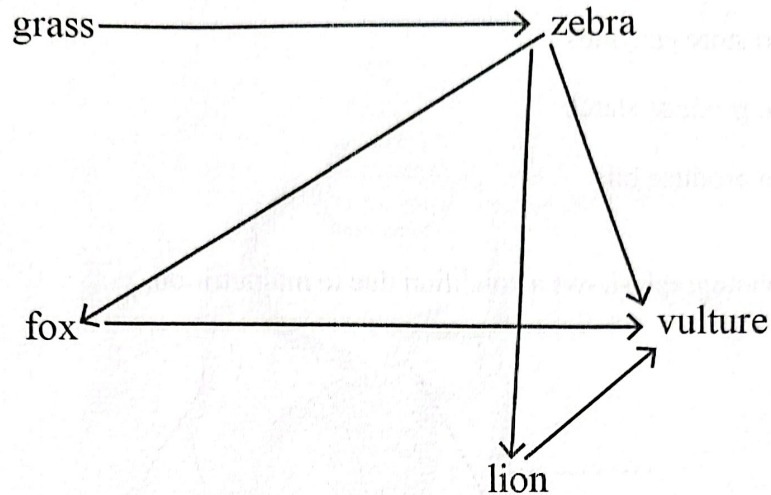
1. The diagram shows a plant cell after it had been placed in a concentrated salt solution.



The plant cell became

- A rigid.
- B turgid.
- C bigger.
- D plasmolysed.

2. The diagram shows part of a food web in Hwange National Park.



The fox is a

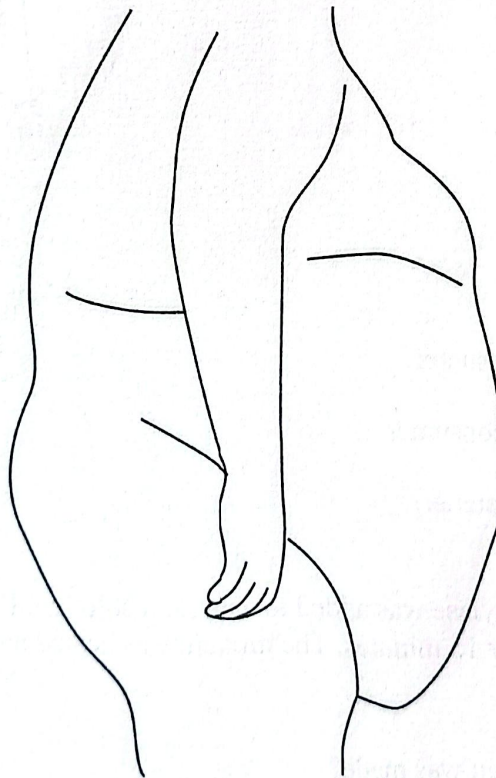
- A producer.
 - B primary consumer.
 - C secondary consumer.
 - D tertiary consumer.
3. The enzyme amylase was added to a nutrient solution. Benedict's solution was added to the mixture after 15 minutes. The mixture was heated and a brick-red colour was observed.

Which conclusion was made?

- A The nutrient solution contained proteins.
- B The nutrient solution contained vitamins.
- C The nutrient solution contained starch.
- D The nutrient solution contained fats.



4. Which one is a function of the liver?
- A to store proteins
 - B to store enzymes
 - C to produce starch
 - D to produce bile
5. The photograph shows a condition due to malnutrition.



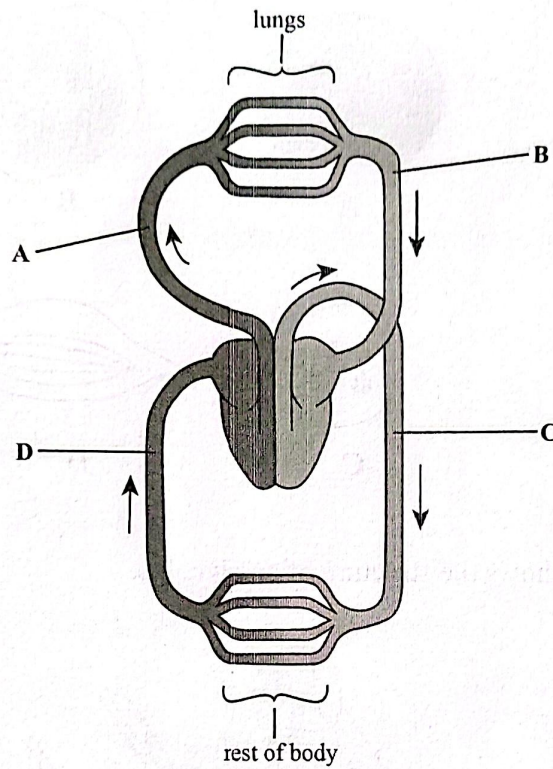
What is a possible cause for the condition?

- A eating excess carbohydrates
- B eating excess fruits
- C eating excess fibre
- D having excessive exercise



6. The diagram shows the human circulatory system.

Which blood vessel, **A**, **B**, **C** or **D**, is the vena cava?

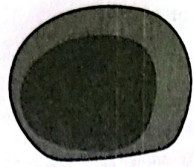
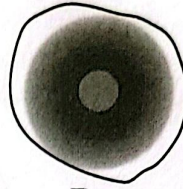
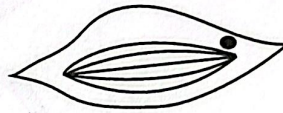


7. Which one is a product of anaerobic respiration in plants?

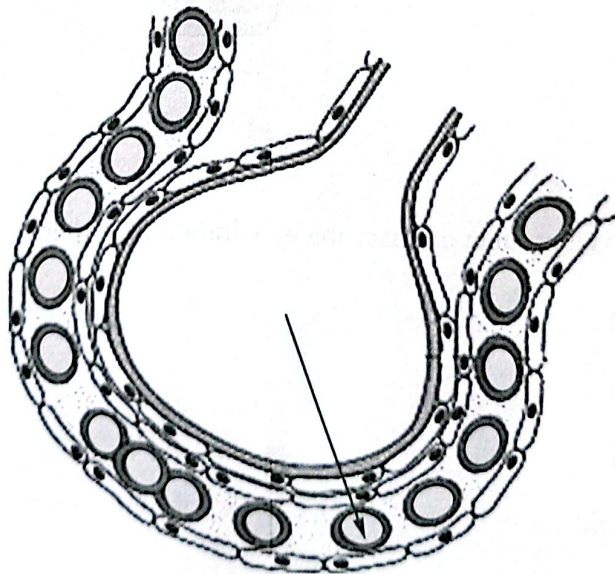
- A oxygen
- B glucose
- C alcohol
- D lactic acid



8. The diagrams show some specialised human cells.
Which cell, A, B, C or D, is a muscle cell?

**A****B****C****D**

9. The diagram shows the structure of an alveolus.

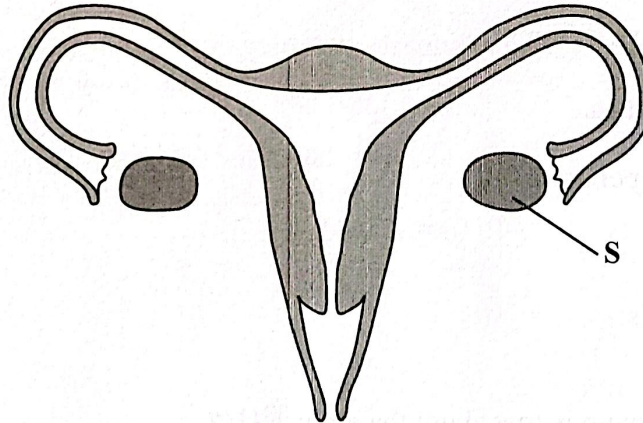


Which process is shown by the direction of the arrow?

- A active uptake
- B respiration
- C osmosis
- D diffusion



10. What is the function of the epididymis?
- A to produce sperms
 - B to store sperms
 - C to produce urine
 - D to store urine
11. Which contraceptive method may also prevent the spread of HIV and AIDS?
- A use of condoms
 - B use of hormonal pills
 - C use of intrauterine devices
 - D use of hormonal injections
12. The diagram shows the reproductive system of a woman.



Which one is a primary function of S?

- A the site for the implantation of the fertilised ovum ✕
- B the site for the fertilisation of the ovum ✓
- C for the production of oestrogen
- D for the production of antibodies ✕



13. A potato is mainly grown from a
- A cutting.
 - B rhizome.
 - C root tuber.
 - D stem tuber.
14. Which disease may be spread by a vector?
- A ebola
 - B malaria
 - C cholera
 - D gonorrhoea
15. Dyes in an ink are separated by paper chromatography.
- The dyes have different
- A solubilities.
 - B boiling points.
 - C colours.
 - D densities.
16. Which statement is true about the atom ${}_{19}^{39}\text{W}$?
- A It has 39 neutrons.
 - B It has 39 protons.
 - C It has 19 electrons.
 - D It has 20 protons.



17. Which elements make Group I of the Periodic Table?
- A noble gases
B halogens
C metals
D non-metals
18. Element K belongs to Group VII.
What is the charge of an ion of K?
- A K^+
B K^-
C K^{7+}
D K^{7-}
19. A sodium hydroxide solution is prepared by dissolving 80.0 g of sodium hydroxide (NaOH) pellets in 2 dm³ of water.
What is the concentration of the solution?
- [Mr of NaOH = 40 g]
- A 1.0 mol/dm³
B 2.0 mol/dm³
C 40.0 mol/dm³
D 160.0 mol/dm³

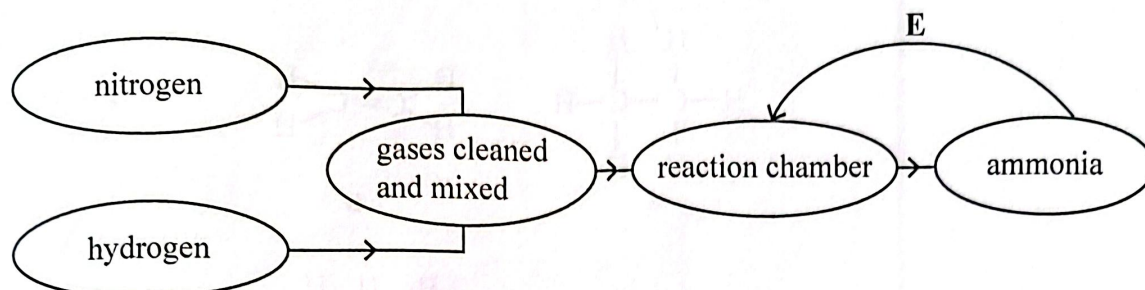
$$C = \frac{m}{v}$$
$$\frac{40,0}{2,0}$$
$$\frac{80,0}{2,0}$$
$$= 40$$



20. Hydrogen gas is a covalent molecule.
Which one is a property of hydrogen gas?
- A It is soluble in water.
 - B It has a high boiling point.
 - C It has a high melting point.
 - D It does not conduct electricity.
21. During electroplating, the object to be electroplated is connected to the
- A cathode.
 - B anode. ✓
 - C electrolyte. ✗
 - D voltmeter. ✗
22. Which reagent is a raw material used to make soap?
- A hydrochloric acid
 - B sodium hydroxide
 - C oxygen gas
 - D nitrogen gas
23. The reaction between hydrochloric acid and sodium oxide produces
- A sodium chloride and water.
 - B sodium hydroxide and water. H
 - C sodium hydroxide and oxygen.
 - D sodium chloride and hydrogen.



24. Nitrogen and hydrogen react to produce ammonia.

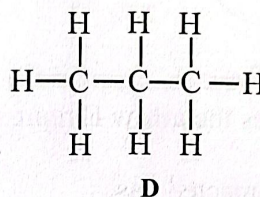
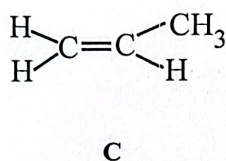
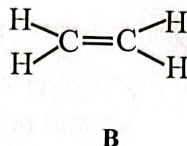
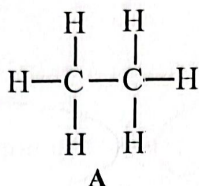


Which process does the arrow E represent?

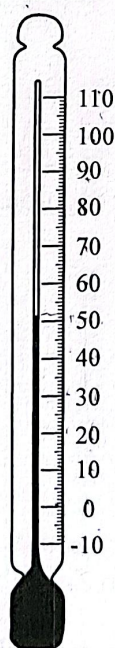
- A cooling of unreacted gases
 - B recycling of unreacted gases
 - C heating of unreacted gases
 - D production of ammonia gas
25. Which one of the metals reacts vigorously with dilute nitric acid?
- A magnesium
 - B copper
 - C zinc
 - D iron
26. Steel is produced at Manhize in Zimbabwe.
- Which mineral ore is processed at Manhize?
- A gold ore
 - B iron ore
 - C copper ore
 - D chrome ore



27. Which organic molecule represents propane?



28. The diagram shows a reading on a thermometer.



What is the temperature shown?

- A 58.5 °C
- B 58.0 °C
- C 51.5 °C
- D 40.5 °C



29. Power is measured in
- A newtons.
 - B joules.
 - C amperes.
 - D watts.
30. A machine used an effort of 60 N to lift a load of 120 N.

What is the mechanical advantage of the machine?

- A 0.5
- B 2.0
- C 60.0
- D 7 200.0

$$MA = \frac{\text{Load}}{\text{Effort}}$$
$$= \frac{120}{60}$$
$$= 2$$

31. What is the advantage of using a solar water geyser over an electric water geyser?
- A the solar water geyser causes air pollution
 - B the solar water geyser uses non-renewable energy
 - C the solar water geyser has low operating costs
 - D the solar water geyser has low energy conversion in winter
32. How is heat mainly transferred in liquids?
- A through conduction
 - B through radiation
 - C through convection
 - D through diffusion

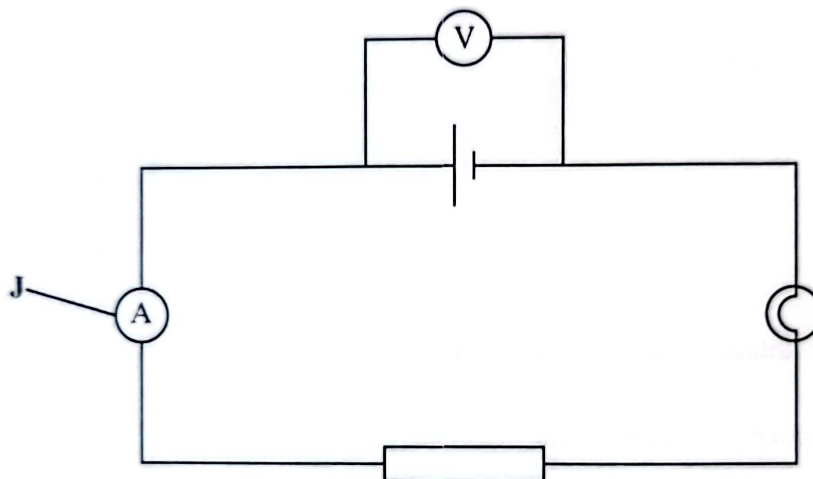


33. A force of 50 N moves a book over a distance of 5 m.

What is the work done?

- A 10 N
- B 45 N
- C 55 N
- D 250 N

34. The diagram shows a simple electric circuit.



What is the function of component J?

- A to measure current ✓
- B to measure voltage ✗
- C to measure resistance
- D to measure light intensity ✓

ZAHAA #11

AKA BABA TOBBY



35. An electric heater is rated 250 V, 1 250 W.

What is the current that is drawn by the heater?

- A 5 A
 - B 1 000 A
 - C 1 500 A
 - D 312 500 A
36. Which one is a safety measure when using electricity at home?
- A overloading adaptors or circuits
 - B using non-insulated electric cords
 - C touching appliances with wet hands
 - D putting on shoes with rubber soles
37. The diagram shows forces acting at a point.



What is the resultant force?

- A 3 N
- B 40 N
- C 80 N
- D 1 200 N



38. Which factor increases the resistance of a wire?
- A decrease in cross-sectional area
 - B decrease in length ✓
 - C decrease in temperature
 - D decrease in mechanical force
39. Why does a vacuum flask have shiny surfaces?
- A to reflect heat ✓
 - B to absorb heat
 - C to conduct heat
 - D to emit heat
40. Which type of energy is primarily converted to other forms of energy during telecommunication?
- A kinetic_x
 - B sound
 - C electrical
 - D mechanical_x
- 55
40
25



9316

